

STATUTORY SURVEILLANCE & DISPATCH RECONCILIATION

SMARTMINEGUARD

Mining & Mineral Transport Monitoring and Anti-Fraud Control System

"Data-driven dispatch reconciliation, GPS surveillance and explainable rule-based enforcement support."

Prepared For:	Mining & Mineral Enforcement Authorities / Directorate of Mines & Geology	Date of Issue:	02 October 2026
Prepared By:	SmartMineGuard Student Development Team (Project Prototype)	Architecture:	Deterministic Software (Zero AI/ML)
Document Purpose:	High-Level Operational & Technical Enforcement Concept Note	Legal Scope:	Decision-Support Tool under MMDR Act Sec 21

EXECUTIVE SUMMARY & PLATFORM OBJECTIVE

SmartMineGuard is a software prototype designed to bring together and authoritatively cross-reference disparate data streams generated across the mineral transportation supply chain. In conventional operations, data is captured in unlinked silos—mine lease records, vehicle registries, driver rosters, e-Rawaana transit permits, AIS-140 GPS feeds, pithead weighbridge receipts, destination receipts, and quarry production stock ledgers. Because these systems rarely talk to one another in real time, fraudulent practices such as transit pass recycling, deliberate overloading, unpermitted route diversions, and clandestine extraction go undetected until audit months later.

Ten Information Streams Integrated & Connected by SmartMineGuard:

- **Mine / Lease Records:** Concession boundaries & limits
 - **Commercial Trucks:** Registration & tare weights
 - **Commercial Drivers:** Transporter profiles & logs
 - **e-Rawaana Permits:** Authorized tonnage & QR hashes
 - **GPS / AIS-140:** Live breadcrumbs & velocity pings
- **Automated Weighbridges:** Pithead gross, tare, net
 - **Spatial Geofences:** Lease boundary & transit buffers
 - **Destinations:** Registered crushers & stockyards
 - **Corridor Checkpoints:** Toll & border monitoring booths
 - **Mass-Balance Ledgers:** Production vs dispatch stock

The system evaluates these records continuously using **deterministic, explainable mathematical and spatial rules**. When anomalies occur, the platform generates instant statutory alerts and compiles structured evidence dossiers for field squads.

CORE STATUTORY PRINCIPLE: "SmartMineGuard is a software-based monitoring and reconciliation platform designed to cross-check mining transportation data and flag potentially irregular mineral trips for officer verification."

Statutory Operational Flow:

SYSTEM DETECTS & FLAGS ■ OFFICER REVIEWS ■ OFFICER VERIFIES ■ ACTION UNDER APPLICABLE RULES

Crucial Notice: SmartMineGuard does NOT claim to independently prove offences or replace human judgment. The enforcement officer remains the sole statutory authority and legal decision-maker under Section 21 of the MMDR Act.

1. WHY MINERAL TRANSPORT REQUIRES CROSS-CHECKING

Commercial mineral transit is an intensive, multi-step operational sequence spanning quarry extraction pits, weighment platforms, state highways, regional checkpoints, and commercial processing facilities. Because each step is managed by different entities using disparate tools (manual gate registers, digital scale indicators, independent GPS dashboards, paper permits), critical operational irregularities frequently slip through unseen.

1. PERMIT	2. TRUCK	3. ENTRY	4. LOADING	5. SCALE	6. DISPATCH	7. GPS	8. RECEIPT
e-Rawaana	Carrier	Geofence	Pithead	Gross/Tare	Gate Exit	Transit	Destination

THE RECONCILIATION IMPERATIVE: "If these events are recorded separately in disconnected registers or standalone databases, irregularities can be exceptionally difficult to identify in real time. SmartMineGuard connects them into one single, auditable trip entity."

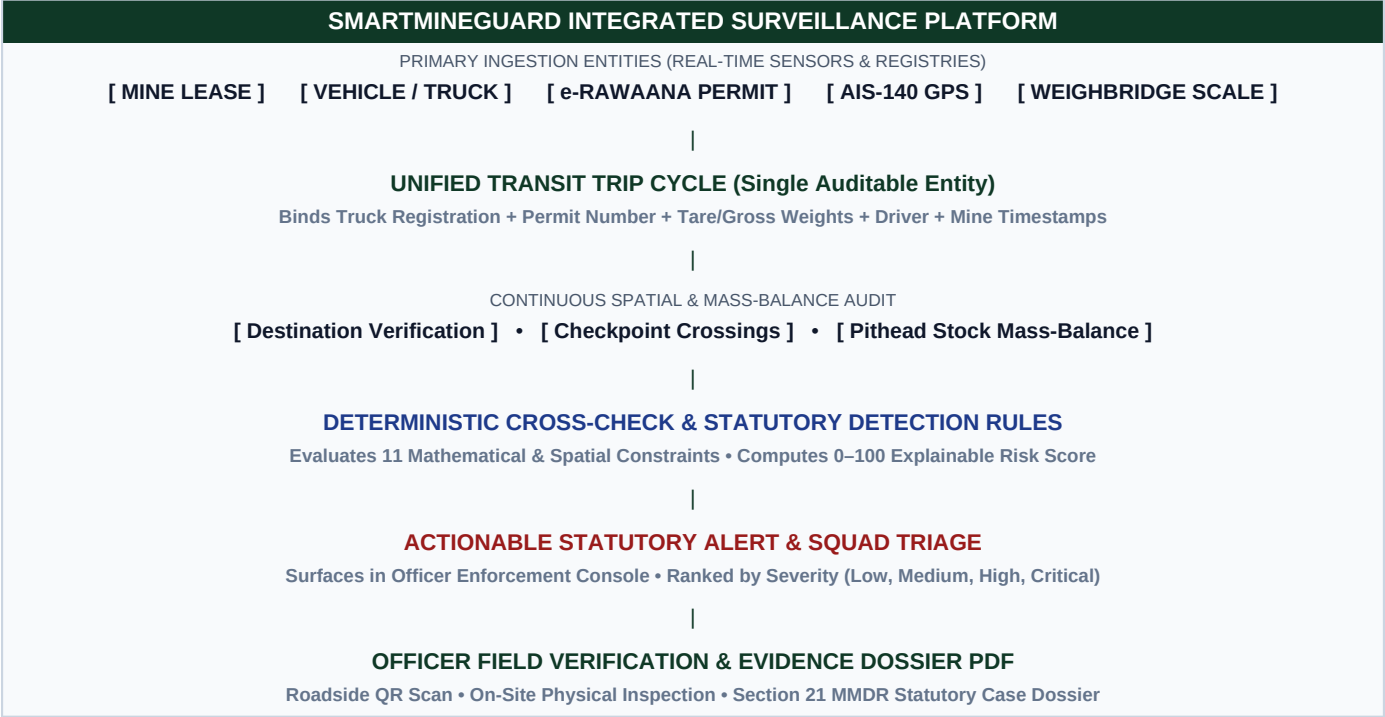
Five Critical Operational Vulnerabilities Addressed:

Operational Vulnerability	How the Irregularity Occurs	Consequence & Enforcement Risk
1. Vehicle Overloading & Tare Inflation	Tipper trucks are loaded beyond authorized e-Rawaana limits or vehicle RTO axle capacity. Drivers tamper with empty tare weight (using false water ballast or off-scale tire parking) to conceal extra cargo.	Extensive road and bridge degradation, evasion of statutory mineral royalty, and severe traffic hazards on commercial highway corridors.
2. e-Rawaana Recycling ("Parchi Reuse")	A legitimate digital transit pass is reused for multiple unbilled trips between quarry and crusher before the expiry window elapses, because delivery confirmation is not synchronized.	Loss of state royalty revenue; multiple truckloads of mineral transit illegally under the legal cover of a single paid pass.
3. Unauthorized Route Diversion	Commercial carriers depart from legal highway transit corridors into unapproved rural links, private crusher bypasses, or prohibited riverbed extraction zones.	Unaccounted dumping, mineral theft, and unmonitored extraction from eco-sensitive buffer zones and forest reservations.
4. GPS Blackouts & Telemetry Tampering	AIS-140 GPS transponders are deliberately disconnected, wire-cut to internal battery, or jammed using portable RF jamming devices while traversing sensitive routes.	Total loss of telematic visibility during active haulage, concealing unpermitted loading or illicit detour routes.
5. Pithead Production vs. Dispatch Mismatch	Quarry leaseholders dispatch mineral tonnages far exceeding their sanctioned annual environmental clearance quotas or report inaccurate stockpile figures.	Unregulated over-extraction, depletion of natural reserves beyond legal concession limits, and fraudulent mass-balance reporting.

The Need for Multi-Source Reconciliation: No single data point—whether a weighment slip or a GPS coordinate—is sufficient to confirm statutory compliance. Only by dynamically binding the vehicle, permit, scale weights, and continuous spatial telemetry into a unified audit trail can enforcement squads detect anomalies with certainty.

2. HOW SMARTMINEGUARD CONNECTS THE DATA

SmartMineGuard transforms disconnected administrative logs into a unified, relational trip entity. Instead of reviewing tables in isolation, the platform links every physical movement to an unbroken digital chain of custody.



The Connected Journey Representation (Conceptual Example):

Each physical haulage movement is modeled as an interconnected chain of certified data points. Consider an illustrative dispatch cycle:

Asset / Parameter	Captured Value	Verification Mechanism & Cross-Check Role
Vehicle Identification	HR26AB1234 (10-Wheeler Tipper)	Cross-referenced against RTO unladen tare weight baselines and registered fleet database.
Statutory Transit Pass	e-Rawaana SMG-2026-00125	Locked to single transit cycle; checks mineral classification, buyer, and active validity window.
Mine Concession & Gate	Aravalli Quarry Block A (08:12 Entry)	Geofence triggers automated mine arrival; matches active permit; initializes trip counter.
Automated Scale Weighment	Gross: 35.8 MT Tare: 11.2 MT	M2M scale capture; derives net payload (24.6 MT); evaluates overload vs. permit (24.0 MT).
Outbound Dispatch Gate	08:42 Mine Exit Clearance	Geofence exit authorizes dispatch; increments daily commercial trip round counter.
GPS Route Tracking	NH-48 Designated Transit Corridor	Calculates perpendicular cross-track deviation from polyline axis in real time.
Destination Delivery	Bhiwadi Crushing Zone	Destination geofence arrival auto-consumes e-Rawaana; locks permit against reuse.

The Enforcement Advantage: By evaluating data as a connected sequence, the system detects discrepancies that are invisible in single-source audits. For example, a truck cannot present a valid weighment slip without an associated mine entry geofence event, nor can it claim transit completion without traversing designated highway checkpoints.

3. FIVE MAJOR CONTROL MECHANISMS

SmartMineGuard enforces statutory compliance through **deterministic, explainable mathematical and spatial rules**. The system strictly avoids opaque AI/ML predictions. Every control mechanism produces a concrete, auditable metric.

CONTROL 1 — AUTOMATED WEIGHBRIDGE CONTROL (Net = Gross - Tare)

In physical operations, net payload weight is calculated directly via automated Machine-to-Machine (M2M) telemetry: **Gross Weight - Tare Weight = Net Mineral Payload**. Operators cannot type or alter the net tonnage manually. If a manual override is required (e.g. scale calibration), it requires authorized supervisor authentication, documented justification, and generates an audit log entry.

- *Illustrative Check:* Gross = 35.8 MT, Tare = 11.2 MT ■ Net = 24.6 MT. Permitted = 24.0 MT. Excess = **+0.6 MT** (+2.5% variance flagged).

CONTROL 2 — GPS GEOFENCING & ROUTE CORRIDOR DEVIATION (>500m Threshold)

Quarry concession perimeters are monitored via spatial geofences that automatically record vehicle arrival and departure timestamps. During highway haulage, real-time GPS breadcrumbs are cross-checked against authorized polyline transit corridors.

- *Documented Rule:* If a transport vehicle deviates more than **500 metres perpendicular distance** from the designated corridor axis, a **ROUTE DEVIATION** alert is raised with exact coordinates and departure distance.

CONTROL 3 — e-RAWAANA RECONCILIATION & PERMIT REUSE PREVENTION

Transit permits follow a strict, irreversible five-stage operational lifecycle: **ISSUED ■ TRUCK ARRIVED ■ WEIGHED ■ DISPATCHED ■ COMPLETED / CONSUMED**.

- *Pass Recycling Prevention:* Each e-Rawaana is cryptographically locked to a single active trip cycle. Once marked CONSUMED at destination, any attempt to initiate a second dispatch on the same permit raises a **CRITICAL PERMIT REUSE** alert.
- *Vehicle-Permit Binding:* Flags an immediate mismatch alert if the physical vehicle registration does not match the permit.

CONTROL 4 — GPS SIGNAL BLACKOUT MONITORING (>15 Minutes Anomaly)

The system continuously audits the heartbeat stream of AIS-140 telematic units during active transit trips.

- *Documented Rule:* If GPS pings cease for **more than 15 consecutive minutes** while a trip is in transit, the platform flags a **GPS BLACKOUT** alert, recording the last known coordinates, heading, and subsequent restoration time.
- *Statutory Clarity:* **A blackout is treated as an operational anomaly requiring officer review and physical verification, not as automatic proof of illegal activity.** (E.g. distinguishing benign mountain cellular dead zones from tampering).

CONTROL 5 — PRODUCTION / DISPATCH MASS-BALANCE RECONCILIATION

At the quarry pithead, extraction stockpiles are governed by the fundamental law of conservation of mass: **Expected Closing Stock = Opening Stock + Today's Production - Today's Dispatched**.

- *Documented Rule:* If physically recorded closing stock differs from expected closing stock beyond the statutory **50 MT configured tolerance**, the system raises a **PRODUCTION-DISPATCH MISMATCH** alert.
- *Illustrative Check:* Opening = 1,250 MT + Blasting = 500 MT - Dispatched = 420 MT ■ Expected = 1,330 MT. If recorded stock is 1,410 MT (80 MT variance), an immediate investigation alert is triggered.

CONTROL INTEGRITY: All 5 controls execute in automated sequence. By removing human manual data entry at scale gates and highway corridors, the platform ensures that records reflect physical reality, giving enforcement squads dependable, verifiable data.

4. FROM DATA TO ACTIONABLE OFFICER ALERT

SmartMineGuard does not generate ambiguous 'AI risk scores' or unexplainable predictions. Instead, every alert follows a transparent, multi-stage processing pipeline grounded in statutory rules.

1. INGESTION M2M Scales & GPS	2. CROSS-CHECK Permit & Geofences	3. RULE TRIGGER 11 Statutory Rules	4. RISK SCORE 0–100 Scale	5. SQUAD ALERT Triage Console	6. VERIFICATION Roadside QR Scan	7. DOSSIER Evidence PDF
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The 11 Deterministic Statutory Detection Rules:

Statutory Rule Name	Concrete Detection Trigger	Risk & Severity	Statutory Objective
1. Weight Anomaly (Overload)	Actual Net Weight > Permitted e-Rawaana Weight (or > vehicle RTO gross limit).	Medium to Critical (+30)	Prevent highway axle damage and recover evaded tonnage royalty.
2. Route Corridor Deviation	Perpendicular cross-track distance > 500m from authorized polyline axis.	Medium to Critical (+20)	Detect unpermitted diversion toward private crushers or riverbeds.
3. GPS Signal Blackout	No periodic AIS-140 telemetry pings received for > 15 consecutive minutes.	Medium to High (+20)	Flag telematic loss; escalate if occurring near sensitive geofences.
4. Permit Recycling / Reuse	Attempted second haul on pass with status marked CONSUMED or EXPIRED.	Critical (+30)	Eliminate 'Parchi reuse' fraud where single passes cover multiple trips.
5. Vehicle-Permit Mismatch	Physical vehicle registration does not match truck ID bound to transit pass.	Critical (+30)	Prevent transfer of permits across unauthorized commercial tippers.
6. Unauthorized Mine Entry	Truck crosses quarry perimeter geofence without a valid active e-Rawaana.	Critical (+30)	Detect unpermitted pithead access and illicit night loading operations.
7. Impossible Speed / Transit	Calculated transit speed > 85 km/h commercial tipper legal limit.	High to Critical (+30)	Expose fake paper-only 'ghost trips' or corrupted GPS telematics.
8. Production Quota Mismatch	Cumulative dispatch > annual quota OR pithead mass balance variance > 50 MT.	High to Critical (+30)	Enforce environmental extraction caps; engage Tender Quota Kill-Switch.
9. Manual Weight Override	Scale gross or tare weight manually altered without automated transducer stream.	Medium (+15)	Audit scale operator interventions; mandate written supervisor reason.
10. Tare Weight Inflation	Recorded empty tare exceeds RTO Vahan baseline by > 1.2 MT and > 8.0%.	High to Critical (+25)	Prevent concealed load fraud where heavy tare conceals mineral cargo.
11. Inter-State Transit (ISTP)	Interstate carrier crosses state border (e.g. Bawal Toll) without valid ISTP pass.	Critical (+35)	Intercept unpermitted cross-border mineral smuggling into Haryana.

TRANSPARENT RISK BANDS: The risk score is capped at 100 points and categorized into transparent bands: **LOW (0–30)**, **MEDIUM (31–60)**, **HIGH (61–80)**, and **CRITICAL (81–100)**. Enforcement officers receive an itemized breakdown showing exactly which rules were triggered and why.

5. ILLUSTRATIVE TRIP AUDIT: ONE TRUCK, COMPLETE TIMELINE

To understand how SmartMineGuard works in practice, review the following documented demonstration scenario. It illustrates how an individual commercial tipper journey is captured as an integrated chronological audit trail.

PROTOTYPE DEMONSTRATION SCENARIO: This scenario is an illustrative prototype demonstration using simulated data to demonstrate system capabilities. It does not represent an actual enforcement action or real private commercial records.			
Target Vehicle Number:	HR26AB1234 (10-Wheeler Tipper)	e-Rawaana Permit No.:	SMG-2026-00125
Mineral Classification:	Quartzite Aggregate (Khanak)	Permitted Transit Weight:	24.00 Metric Tonnes (MT)
Scale Gross Weight:	35.80 Metric Tonnes (MT)	Scale Tare Weight:	11.20 Metric Tonnes (MT)
Calculated Net Payload:	24.60 Metric Tonnes (MT)	Net Variance / Excess:	+0.60 MT Overload (+2.5%)

Complete Chronological Trip Milestone Ledger:

Time	Milestone Event	System Action & Statutory Audit Finding	Status / Risk
08:12	Mine Geofence Entry	Truck crosses quarry perimeter. GPS triggers automated arrival; searches active permit registry.	NORMAL (0)
08:15	Permit Matched	Valid e-Rawaana SMG-2026-00125 auto-matched. Quota checked; shift round counter initialized.	VALID (0)
08:22	Pit Loading Started	Truck positions at quarry extraction hopper. Trip status transitions to LOADING.	NORMAL (0)
08:28	Scale Weighment	Pithead scale captures Gross (35.8 MT) and Tare (11.2 MT). Platform IR interlock confirms deck alignment.	WEIGHED
08:29	Net Payload Audit	Deterministic calculation: Net = 24.6 MT vs. Permitted = 24.0 MT. Excess +0.6 MT logged in ledger.	OVERWEIGHT (+30)
08:34	Dispatch Cleared	Outbound transit pass generated. Weighment manifest sealed cryptographically.	DISPATCHED
08:42	Mine Geofence Exit	Vehicle exits mine perimeter geofence. System starts transit timer; increments driver round tally.	IN_TRANSIT
09:25	GPS Blackout Event	AIS-140 telemetry pings cease along transit corridor. 15-minute timer triggers alert to squad console.	BLACKOUT (+20)
10:05	Signal Restored	Telemetry resumes near highway junction. 40-minute blackout window archived as investigative evidence.	FLAGGED (50)

Why Chronological Timelines Matter to Enforcement Officers:

In traditional enforcement, an officer intercepting truck HR26AB1234 at 10:30 AM would see only a printed paper permit and have no way to know that the truck experienced a 40-minute telemetry blackout or was dispatched with an unbilled +0.6 MT overload. SmartMineGuard gives the officer the **entire unbroken lifecycle** at a glance.

6. FROM ALERT TO OFFICER FIELD VERIFICATION

SmartMineGuard is engineered as an officer decision-support platform. The software automates surveillance, data correlation, and evidence compilation—empowering field squads to conduct rapid, legally sound verifications.

The 10-Step Operational Enforcement Workflow:

Stage	Officer Action & System Interaction	Operational Outcome
Step 1	Alert Surfaces in Console: An alert appears in the Enforcement Squad queue ranked by severity.	Squad prioritizes target vehicle based on risk score.
Step 2	Select Vehicle: Officer selects vehicle (e.g. HR26AB1234) to open the real-time inspector.	Displays active permit, driver, and current coordinates.
Step 3	Review Flagged Reason: Officer inspects exact statutory rule triggered and numerical variance.	No ambiguous predictions; concrete violation data.
Step 4	Live GIS Tracking: Officer views vehicle on Leaflet GIS map with route corridor buffers.	Tracks vehicle trajectory toward checkpoint intercept.
Step 5	Inspect Timeline: Officer reviews the chronological event ledger from pithead to transit.	Identifies blackout duration or corridor departure point.
Step 6	Roadside QR Scan: Intercepting squad scans vehicle's digital/paper QR code via mobile camera.	Instant verification: Confirms valid, expired, or recycled pass.
Step 7	Review Weighment: Officer examines certified scale gross, tare, and net weighment receipt.	Confirms whether weight exceeds registered gross capacity.
Step 8	Examine Telematics: Officer reviews spatial deviation distance (meters) or blackout log.	Corroborates driver explanation against GPS data.
Step 9	Record Field Findings: Officer enters on-site physical inspection notes into mobile portal.	Immutable audit trail created under officer badge ID.
Step 10	Generate Evidence Dossier: System compiles structured case record into an official PDF dossier.	Official documentation ready for statutory compounding.

Contents of the Statutory Evidence Dossier PDF:

- When an officer elevates an alert to a formal investigation file, SmartMineGuard compiles an official dossier containing:
- **Vehicle & Owner Profile:** Registration number, chassis classification, registered transporter, and RTO tare baseline.
 - **e-Rawaana Permit Metadata:** Pass number, QR security hash, sanctioned tonnage, mineral grade, and destination.
 - **Certified Scale Receipts:** Gross, tare, and net weights with automated scale ID and manual override audit history.
 - **Chronological Milestone Log:** Exact timestamps of mine entry, loading, scale capture, gate exit, and checkpoints.
 - **Spatial Deviation Evidence:** Measured perpendicular departure distance (meters) and GPS telemetry blackout timestamps.
 - **Officer Notes & Findings:** Field inspection observations, compounding fee calculations, and official signature blocks.

STATUTORY & LEGAL INTEGRITY NOTICE: "The Evidence Dossier is designed to support official administrative investigation and statutory documentation. It does not replace statutory administrative procedures or judicial discretion under Section 21 of the MMDR Act."

All compounding orders, vehicle seizures, and formal penal notices remain subject to the physical inspection, substantive findings, and final determination of the authorized mining officer.

7. PROTOTYPE SCOPE & DEPLOYMENT PATH

In the spirit of technical transparency and academic rigor, the current operational boundaries of the SmartMineGuard demonstration prototype are explicitly documented below:

System Dimension	Current Demonstration Prototype State	Production Deployment Requirement
GPS Vehicle Tracking	Simulated multi-truck telematic movement across realistic Haryana mining corridors.	Direct server webhook / MQTT ingestion from physical AIS-140 GPS transponder vendors.
Weighbridge Scale Feeds	Structured synthetic weighment datasets representing real-world Gross/Tare values.	Standard RS-232 / WebSerial / IoT digital indicator bridges on physical scale decks.
National Portals (VAHAN)	Simulated RTO unladen tare baselines and vehicle ownership registries in local DB.	Secure API integration with National Transport Project (VAHAN/SARATHI) gateways.
State e-Rawaana Integration	Embedded statutory permit generation and QR verification mimicking state rules.	Bi-directional REST API integration with State Department e-Rawaana production server.

The Six-Stage Roadmap to Operational Deployment:

STAGE 1: Functional Prototype & Academic Validation (Current Status: Fully Operational Demonstration)
STAGE 2: Technical & Statutory Validation with Mining Department & Enforcement Squad Officials
STAGE 3: Field Squad & Domain Expert Feedback to Refine Regional Thresholds & Geofence Coordinates
STAGE 4: Controlled Pilot Deployment on a Single Selected Quarry Corridor / Weighbridge Cluster (e.g. Khanak)
STAGE 5: Live API Integration with AIS-140 Telematic Gateways & Certified Weighbridge Scale Transducers
STAGE 6: Statewide Operational Deployment & Ongoing Statutory Surveillance Rollout

INVITATION FOR TECHNICAL EVALUATION: "SmartMineGuard is respectfully presented for technical and operational feedback, domain evaluation, and statutory guidance from competent Mining & Enforcement Authorities."

Project Title:	SmartMineGuard	Target Sector:	Mineral Transit & Concession Surveillance
Core Repository:	https://github.com/jaipalverma808/smartmineguard	Demonstration:	Standalone Software Prototype (Flask/PostGIS)
Development Team:	SmartMineGuard Student Development Team (Team Codeavengerz)	Submission Track:	Smart India Hackathon (SIH) 2026